mail-agent

> Outlook Graph → → Send
> .md .md

0.

`customer.care@ismart-service.com` Inbox → / / → **Drafts**
\+ → **Drafts** / **Send** `sendMail`

1.

`sendMail` `createReply` → Drafts
/ move Graph DELETE / move `isRead`
/ `en-US`
SLA prompt + “1-2 business days”
PII git `~/Documents/private/iSmart-email-ledger/`

`python3 -m agent.outlook_drafts agent/outlook_drafts.py`

---

2. AI

2.1

	Python 3.11+	
LLM SDK	**openai** Azure OpenAI	Chat Completions + Structured Outputs
	Pydantic v2	Intent / Triage / Draft / Compliance schema
Token	**tiktoken** `o200k_base`	
	python-dotenv	`.env` Azure AD / OpenAI / MAILBOX_*
	openpyxl	`iSMART` `_` `_*.xlsx`
API	**Microsoft Graph** + **msal** + **requests**	Inbox createReply PATCH
	Vision / **pypdf** / python-docx	PDF/Word

`requirements.txt` Qdrant FastAPI Celery

2.2 LLM

	Azure OpenAI Cognitive Services
/	**gpt-5.4** `model_name` = `deployment` = `gpt-5.4`
	`agent/azure_llm.py` → `AzureLLMClient`
	`invoke_json(schema=PydanticModel, messages=..., reasoning_effort=...)`
	OpenAI **Structured Outputs** `response_format=json_schema` + `strict=true` Pydantic schema
	`reasoning_effort` `low` / `medium` / `high`
Prompt Cache	`ROLE_AND_LAWS_*` / Azure cache `cached_tokens`

→ vision triage `is_ismart_app` +

`AZURE_OPENAI_ENDPOINT` `AZURE_OPENAI_API_KEY` `AZURE_OPENAI_API_VERSION` gpt-5.4

Outlook **LLM**

`role=`	**reasoning**	**Schema /**	**Prompt**
`triage`	low	`TriageResult` 38 Intent urgency region red_flags...	`TRIAGE_SYSTEM`
`vision_triage`	low	JSON	`_VISION_EXTRACT_PROMPT` outlook_drafts
`template_pick`	low	`TemplatePick.template_id`	`TEMPLATE_PICK_SYSTEM`
`semantic_review`	low		`SEMANTIC_DRAFT_REVIEW_SYSTEM`

LLM + / LLM

/ **Agent** `agent/email_agent.py` Drafts

	reasoning	**Prompt**
`retrieval`	high	`RETRIEVER_SYSTEM`
`drafting` / `rewrite`	high	`DRAFTING_SYSTEM` / `REWRITE_SYSTEM`
`compliance`	high	`COMPLIANCE_SYSTEM` + `ROLE_AND_LAWS_*`
`reflection`	high	`REFLECTION_SYSTEM`

2.3 Prompt

`agent/prompts.py` git diff / A-B /

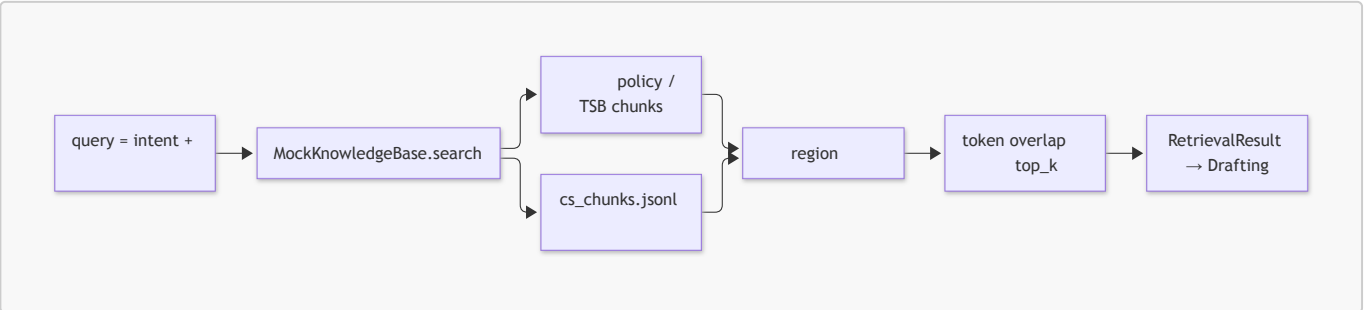
`ROLE_AND_LAWS_NA` / `_EU` / `_UK` / `_OC` /	LATAM·MEA·APAC +	cache
`COMPLIANCE_CONSTRAINTS_*`		
`TRIAGE_SYSTEM`	32 Intent rubric auto_reply_level	urgency
`TEMPLATE_PICK_SYSTEM`		`template_id`
`SEMANTIC_DRAFT_REVIEW_SYSTEM`		
`RETRIEVER_SYSTEM` / `DRAFTING_SYSTEM` / `COMPLIANCE_SYSTEM` / `REFLECTION_SYSTEM`	Agent	

- prompt cache
- **prompt**—— prompt §9
- user / drafting SLA ——
  + TEMPLATE_PICK /

2.4 Excel

`iSMART`	`_`	`_YYYYMMDD.xlsx`
`load_xlsx()`		`template_id`
ID		
list[dict]	/	`template_id`
CS	→ v4	/ CS chunks
	+	RAG

2.5 RAG —



	agent/tools/mock_rag.py	**Qdrant**
	token overlap embedding intent _	
Policy chunks	/	
CS chunks	~/Documents/private/…/cs_chunks.jsonl scripts/build_cs_chunks.py outbox	
Region	chunk.metadata.region enum North America NA	
	EmailAgent **/ eval/run_eval.py**	
Outlook	**RAG** Excel **TCMP / CRM** MockKnowledgeBase	

2.6 LLM

/	template_reply.py	maintenance history app not responding
_review_lock /		template_id GEN-001/BIND-001
/	outlook_drafts.py	customer_issue_resolved pure_chase_nudge we_replied_last
Region /	region_catalog.py	Europe+GB → United Kingdom
Graph	GraphClient._request	DELETE / sendMail / PATCH
	mail_tags.py	VIN / / L2

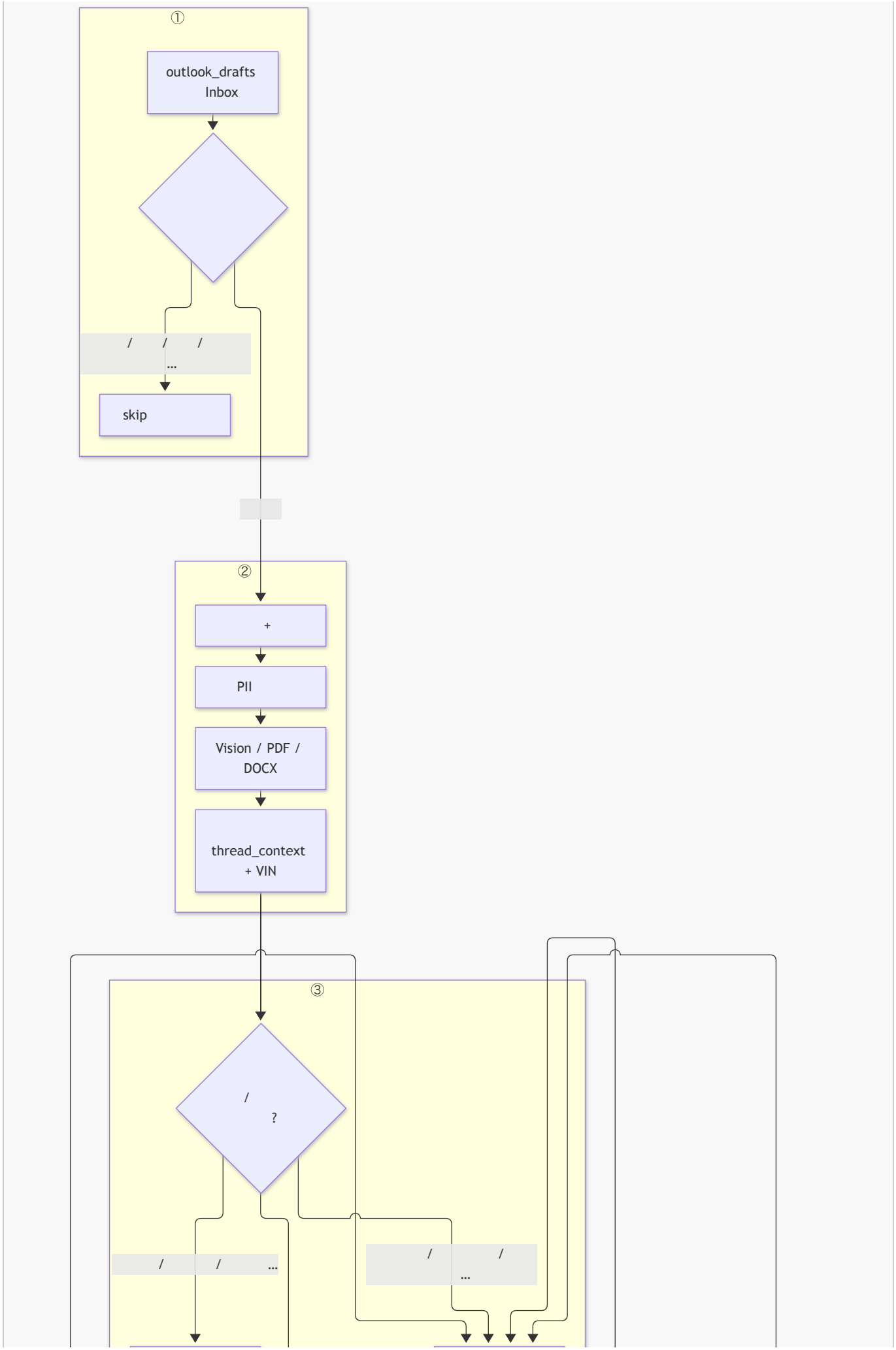
2.7

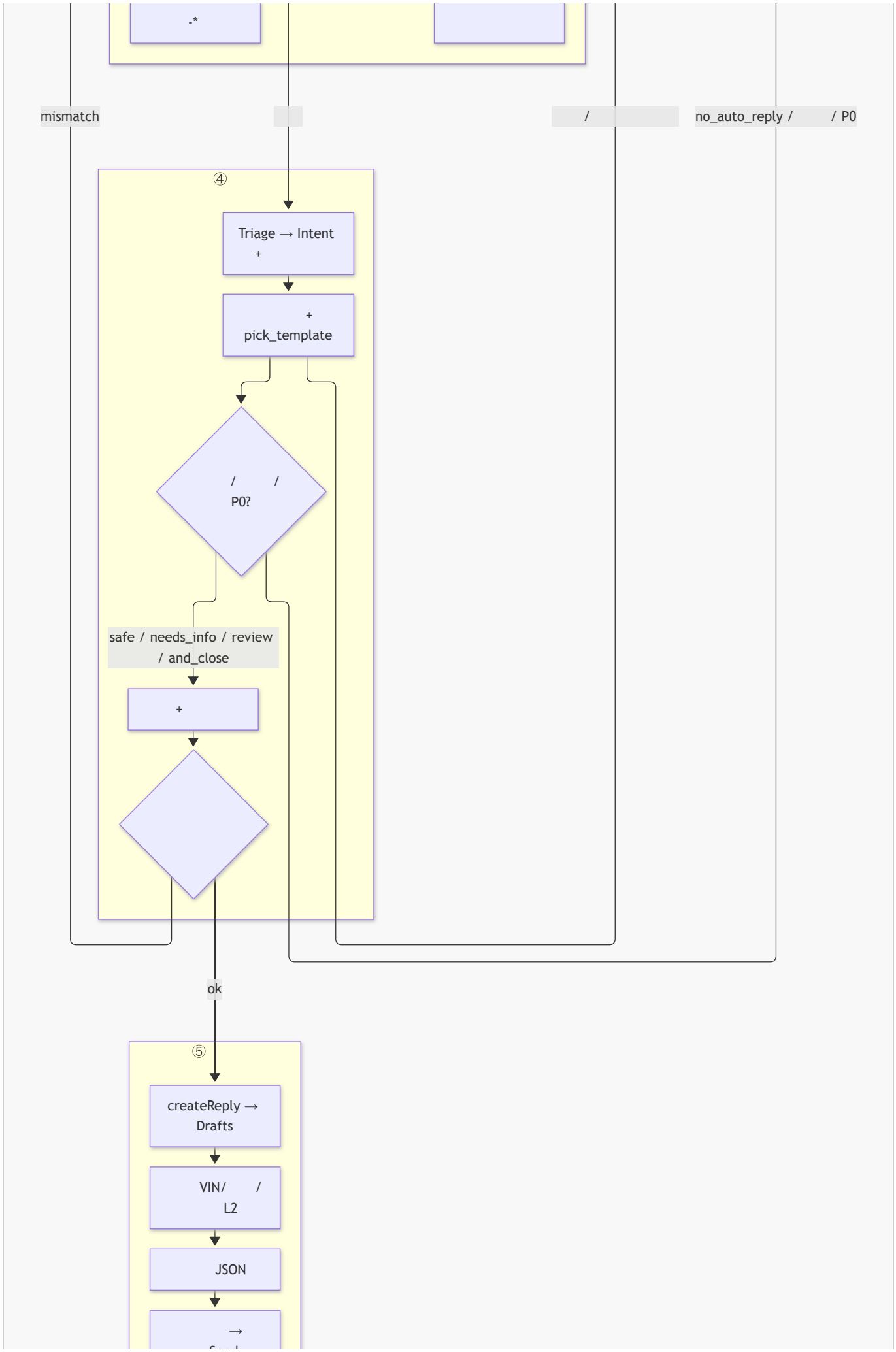
		AI	
	outlook_drafts → run_template_reply	Triage + Vision + Template pick +	Outlook Drafts
/ PoC	email_agent.EmailAgent	Triage → Mock CRM → RAG → Draft → Compliance → Reflection	eval JSON / send

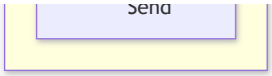
3.

Outlook Drafts 5

①	Graph Inbox			msal / Graph	
②	+ Vision/PDF +			redact / gpt-5.4	
③	/ / ...				LLM
④	**Triage Intent** →	→	→	LLM + Excel KB +	
⑤	createReply +	+		Graph	


①
outlook_drafts
Inbox
/ / /
...
skip
②
+
PII
Vision / PDF /
DOCX
thread_context
+ VIN
③
/
?
/ / ...
/ /
...


-*
mismatch
/
no_auto_reply /
/ P0
④
Triage → Intent
+
+
pick_template
/
/
P0?
safe / needs_info / review
/ and_close
+
ok
⑤
createReply →
Drafts
VIN/
/
L2
JSON
→



- **Triage** Intent / urgency / region / red_flags...
- `pick_template` LLM `template_id`
- / Drafts

Triage **§6.1** **§6.3**

4.

4.1

```
#
python3 -m agent.outlook_drafts --once --top 100 --include-read \
  --since 2026-09-15T16:00:00Z --until 2026-09-16T16:00:00Z

#
python3 -m agent.outlook_drafts --every 15 --top 100

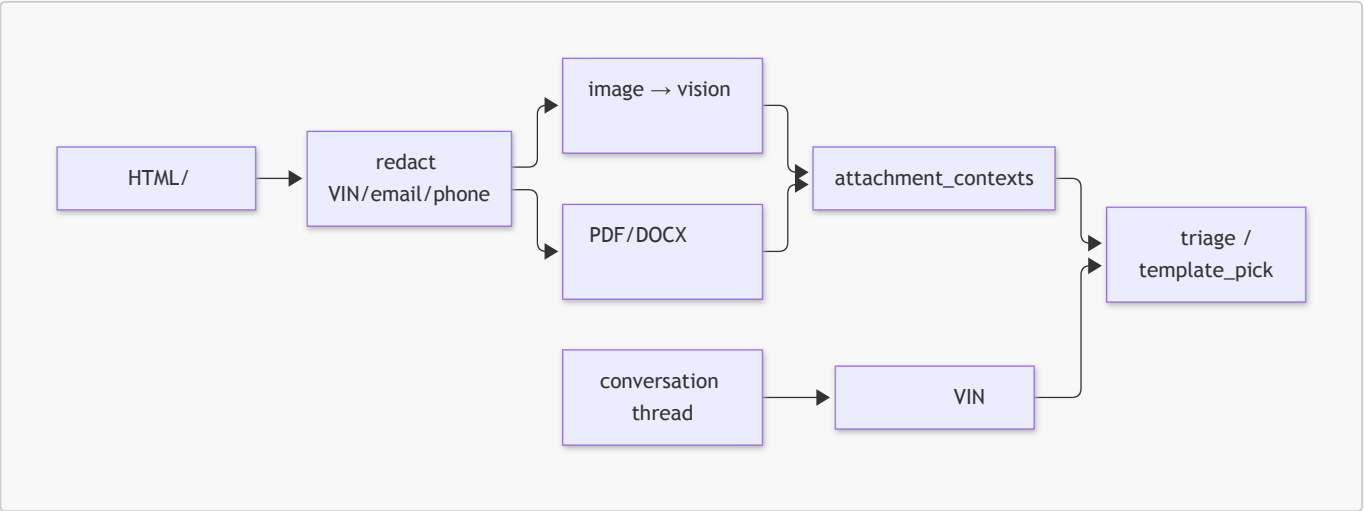
#       +         /
python3 -m agent.outlook_drafts --message-ids '...' --replace-existing-draft
```

- Graph `Mail.ReadWrite` Delegated App token `~/.cache/emailagent/`
- `outlook_drafts.lock`
- `~/Documents/private/iSmart-email-ledger/outlook_scan_*.json`

4.2 LLM

– / –
Office /
/ Legalization –
`mail_id` `already_scanned` `allow_rescan`
`we_replied_last`
`already_has_draft` `--replace-existing-draft`
`follow_up_needs_human` →
/ `dealer_unresolved_needs_human` →

5.



- Vision
-
- screenshot

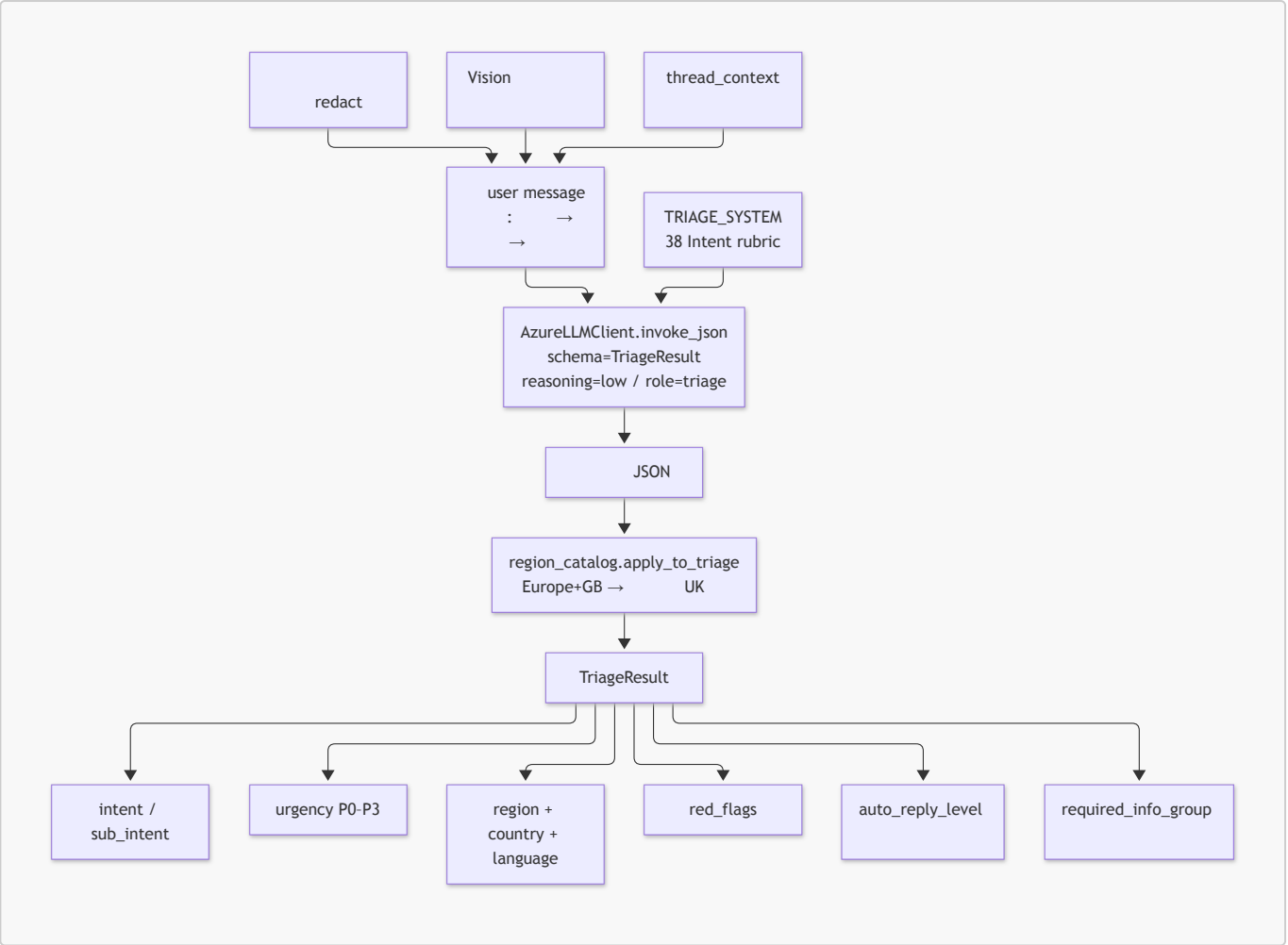
6.

6.1 Triage Intent —

32 Intent xlsx 32 schema / prompt **38** 32 base + M2.5 6 `ficm_bluetooth` / `ficm_voice` / `ficm_radio` / `ismart_translation` / `ismart_music_streaming` / `tsp_data_missing` enum `schemas.py` + `TRIAGE_SYSTEM` +

`template_reply.classify_intent()` ← `run_template_reply()` ← `outlook_drafts.process_mail()`

6.1.1



1. webform → Intent
2. **Vision** `--- Attached screenshot descriptions ---` → / system prompt cache
3. → vs follow-up

System	`prompts.TRIAGE_SYSTEM` ROLE + + 38 Intent / urgency / red_flag / auto_reply_level rubric
Schema	`schemas.TriageResult` → OpenAI Structured Outputs `strict`
reasoning_effort	`low`
	`--no-llm` / client `_keyword_classify` smoke LLM

6.1.2

`intent`	38	`pick_template(triage.intent, …)` GEN	hint
`sub_intent`	L2 67		`sub_intent_unknown`
`urgency`	P0–P3	P0 →	ADAS P1
`region` / `country`	+	region gate /	picker `apply_to_triage`
`language`			
`red_flags`	/ / /	→	auto_reply_level
`auto_reply_level`	5		**xlsx**
`required_info_group`			xlsx please provide
`requires_*_lookup`	/	Agent / Mock CRM CRM	Outlook

`lawyer_letter` / `litigation_threat` / `media_cc` / `social_media_threat` / `safety_incident`

5 `auto_reply_level`

`auto_draft_safe`
`auto_draft_needs_info`
`auto_draft_review`
`auto_draft_and_close`
`no_auto_reply` →

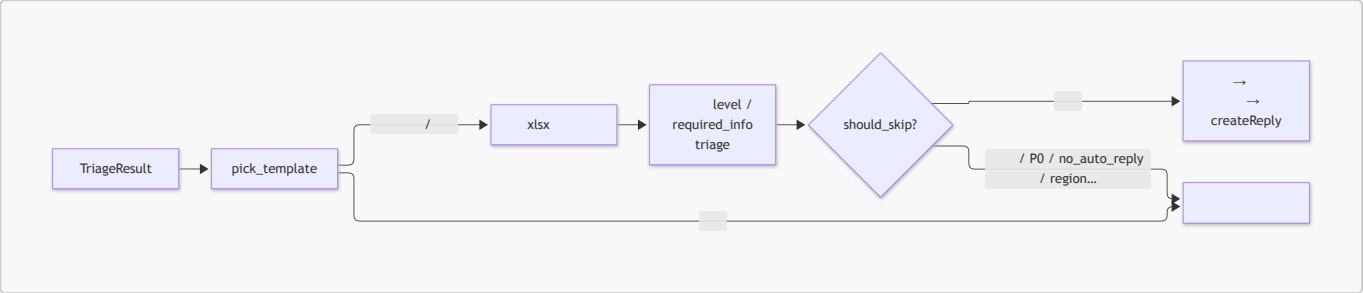
6.1.3 Intent 38

	Intent
App	`app_vehicle_control` `app_account` `app_binding` `app_update` `app_messages` `app_suggestion`
iSMART	`ismart_data_plan` `ismart_remote_control` `ismart_digital_key` `ismart_fota` `ismart_connectivity` `ismart_account_inquiry` `ismart_owner_change` `ismart_region_support` `ismart_privacy` `ismart_user_manual` `ismart_feature_inquiry` `ismart_translation` `ismart_music_streaming`

	Intent
FICM	`ficm_activation` `ficm_cpaa` `ficm_map` `ficm_bluetooth` `ficm_voice` `ficm_radio`
TSP	`tsp_sensitive_data` `tsp_data_missing`
ADAS	`adas_acc` `adas_aeb` `adas_lka` `adas_tja`
/ iSMART	`hardware_warranty` `charging` `delivery_handover` `vehicle_incompatible`
	`complaint` `follow_up_resolved` `other`

`TRIAGE_SYSTEM` CarPlay ≠ `ficm_cpaa` GPS vs App TSP Intent

6.1.4 Triage



- Triage `STATUS-003` ID `TEMPLATE_PICK_SYSTEM` +
- ≠ **Triage prompt** §1.3
- `--triage-only` Intent/urgency region gate / /

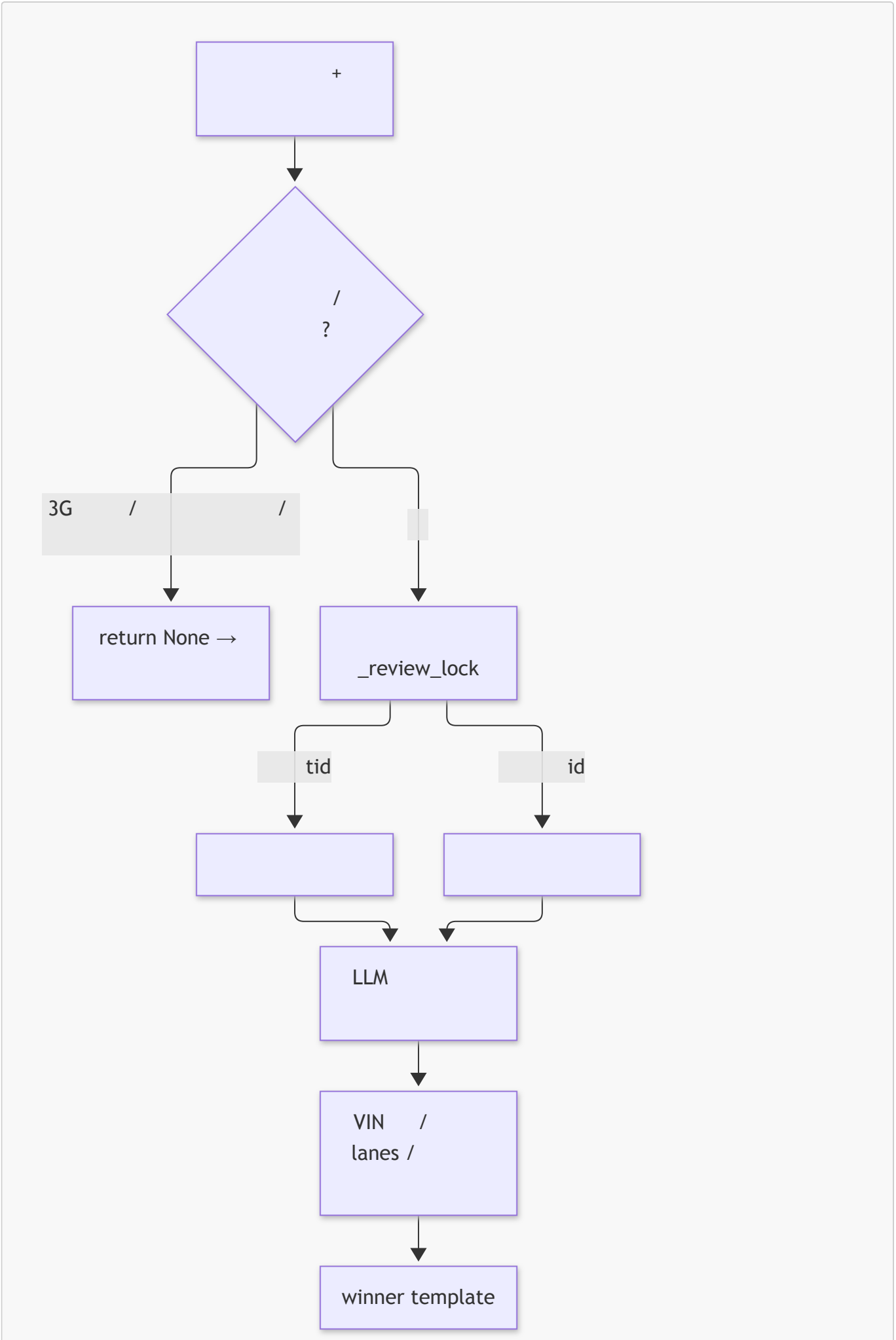
6.2

```
iSMART              _       _YYYYMMDD.xlsx
  └─ load_xlsx()                                  template_id
```

`template_id` `auto_reply_level`

`…_20260918.xlsx`

6.3 `pick_template`


+
/
?
3G / /
return None →
_review_lock
tid
id
LLM
VIN /
lanes /
winner template

.md §1.3

1. /
2. None →
3. template_id +
4. TEMPLATE_PICK_SYSTEM + eval/test_xlsx_scenario_routing.py

detect_issue_lanes → compose_merged_template

6.4

level
auto_draft_safe
auto_draft_needs_info VIN/
auto_draft_review
no_auto_reply

draft_semantic_mismatch_reason

6.5 Region gate

- **1** / GDPR →
- **2** / / REG-001
- M2_ROLLBACK_TO_M1=1 NA

7.

7.1

1. {{salutation}} /
2. HTML + Drafts
3. POST .../messages/{id}/createReply → Drafts
4. PDF

7.2 move

:… / L2 L2
VIN:… / :… / Region:…
– / –

Outlook

7.3

JSON `mail_id` `skipped` / `draft_id` `template_id` `intent` vision `fact_tags`

`~/Documents/private/iSmart-email-ledger/` **git**

8.

`agent/outlook_drafts.py`	Graph `process_mail` Vision
`agent/azure_llm.py`	Azure OpenAI gpt-5.4 `invoke_json` / Structured Outputs / usage
`agent/template_reply.py`	
`agent/prompts.py`	system prompt Triage / Pick / Compliance / ...
`agent/schemas.py`	Intent / level / Pydantic
`agent/mail_tags.py`	VIN/ / L2 PATCH categories
`agent/email_agent.py`	Agent Triage→RAG→Draft→Compliance + region gate
`agent/tools/mock_rag.py`	Mock + CS chunks token overlap Qdrant
`agent/greeting.py`	Dear Mr./Ms.
`eval/test_xlsx_scenario_routing.py`	/
`eval/test_outlook_safety.py`	Graph send/delete/move
`eval/run_eval.py`	triage-only / full eval

9.

prompt

1. `.md` §1.3 → tid /
2. `template_reply.py` + `_review_lock` /
3. `_specialize_template_for_ask` id
4. `prompts.py` `TEMPLATE_PICK_SYSTEM`
5. `eval/test_xlsx_scenario_routing.py` Fxx
6. `python3 eval/test_xlsx_scenario_routing.py`

Intent enum `schemas.py` + `prompts.py` +

10.

```
#
python3 -c "from agent.template_reply import load_xlsx, ROOT; \
 from pathlib import Path; \
 kb=sorted(ROOT.glob('iSMART              _       _*.xlsx'), reverse=True); \
 print(kb[0].name); print(len(load_xlsx()))"

#
python3 eval/test_outlook_safety.py

#
python3 eval/test_xlsx_scenario_routing.py

#
python3 -m agent.outlook_drafts --stop
```

11.

Web	Outlook Drafts			
Send / 72h	sendMail		safe	
CRM / RAG	TCMP	RAG	Mock+CS chunks	§2.5

23h		23	.md

Drafts